

An abstract graphic in the top right corner featuring a vibrant splash of colors including blue, purple, pink, and red. The splash has a textured, dotted appearance, resembling ink or paint splatters.

Web Design Principles

BIS605

Software Development Technologies for Digital Business

Asst. Prof. Dr. Worarat Krathu

Resources

- Antonio De Pasquale, Responsive Web Design, Responsive Web Design Workshop, Milan March 2014
- Mark Branom, CS 21 – Creating and Building Websites, <https://web.stanford.edu/group/csp/cs21/>
- <https://webdesign.tutsplus.com/articles/an-introduction-to-color-theory-for-web-designers--webdesign-1437>
- <https://blog.one.com/colortheory2/>
- <https://99designs.com/blog/creative-inspiration/psychology-color-web-design/>
- <https://websitesetup.org/website-color-schemes/>
- <http://mmicreative.com/the-60-30-10-rule/>
- Materials of BIS605 lectures

Outline

- Things must concern
- Responsiveness
- Design principles
 - Layout & Alignment
 - Golden section
 - 12 grids
 - Color
 - 3 tiers color
 - Mood/tone
 - Pages organization

The Making of a Good Design

- Content is important, but content alone will not make your site work.
- Good Design is:
 - Understandable
 - Interesting
 - Easy to use
 - Uniform in look and feel
 - Done from a visitor's point of view:
WYSIWYW (What You See Is What You WANT)

Good Design Maxims

- “Rules” are only guidelines -- no single model fits every situation, and there is no such thing as the “right” way to create a web site.
- Remember WYSIWYW
 - Web users want control over the online material -- they want to seamlessly obtain the information they need.
 - Don’t force visitors down a specific path -- give them control.

Typical Website Evolution

Generation 1 -- replaces paper information

Generation 2 -- has flashy elements

Generation 3 -- is bleeding edge, causing content to suffer

Generation 4 -- content and technology are integrated

- Ideally, try to skip the problems of Generations 1-3 by planning your web site carefully.

Influences of Technology on Design

- Browsers
- Operating systems
- Connection speeds
- User screen sizes

Influences of Content on Design

- The content drives how the web site looks
 - Sites designed for students look different than sites designed for staff, which look different from sites designed for potential faculty
 - Sites designed for current employees look different than sites designed for potential clients
 - Sites designed to get people to purchase items look different than sites designed to provide information
- Use quality content from subject matter experts
- Have site reviewed PERIODICALLY by key members (CEOs, Department Heads, Supervisors, etc.) of the group the site supports
- Have non-affiliated people review content for clarity
- Have others proofread for grammar

Fresh eyes often see things you miss!

Usability

- Browsers don't use web sites -- people do. Don't design a site for a particular browser -- design a site for the user.
- Make it easy to read
- Ease user's memory
- Care for response time

Arbor Business Company, INC.

President: John Smith

VP Marketing: Susan Jones

Alice Johnson, Manager

Tim Moore, Manager

VP Sales: Rachel Parker

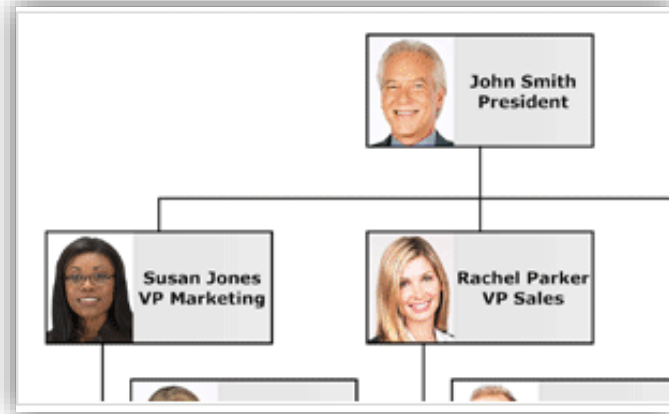
Michael Gross, Manager

Kim Dole, Manager

VP Production: Tom Allen

Kathy Roberts, Manager

Betsey Foster, Manager



Top 3 Website Design Mistakes

- Designing your website for your needs, not the needs of your visitors.
- Designing your site in a way that people can't figure out what they're supposed to do.
- Font & Color
 - Don't use fonts or content that makes it difficult to read
 - Don't make the font size any smaller than 80% of the default size for the browser.
 - Don't use graphics as text
 - Don't use dense text
 - Don't do "mystery meat navigation"

RESPONSIVENESS

What is the Shape of Digital Experience?

- Same contents, different shapes



Credit: <https://mobilepunditsblog.wordpress.com/>

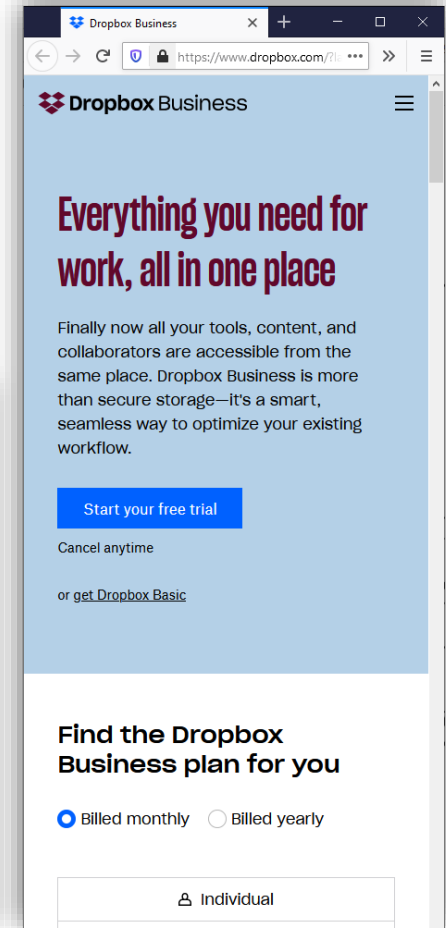
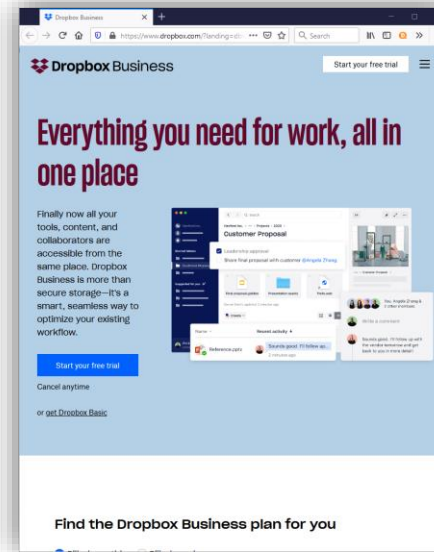
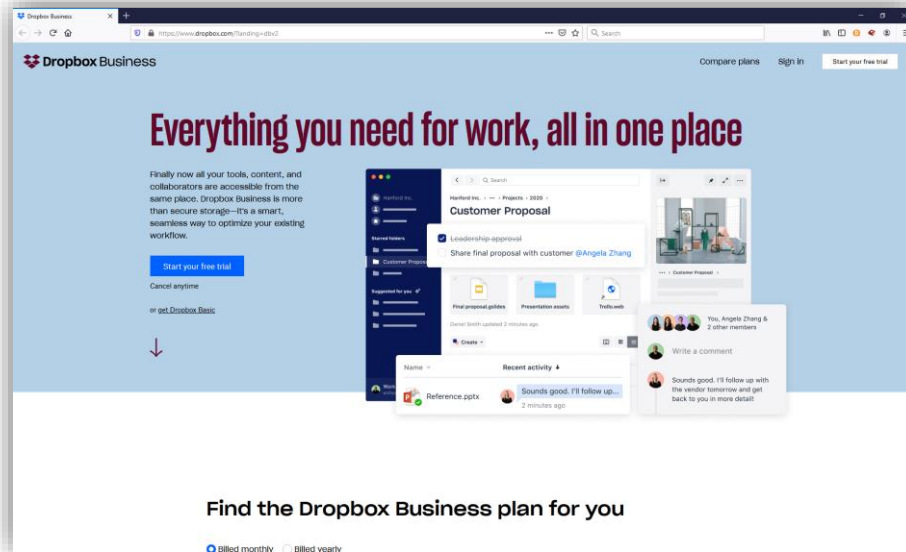
Credit: <https://go.harveynorman.ie/technology/7167-tablets-buying-guide>

Credit: <https://www.apple.com/th/shop>

Credit: <https://th.priceprice.com/Xiaomi-Mi-Gaming-Laptop-25833/>

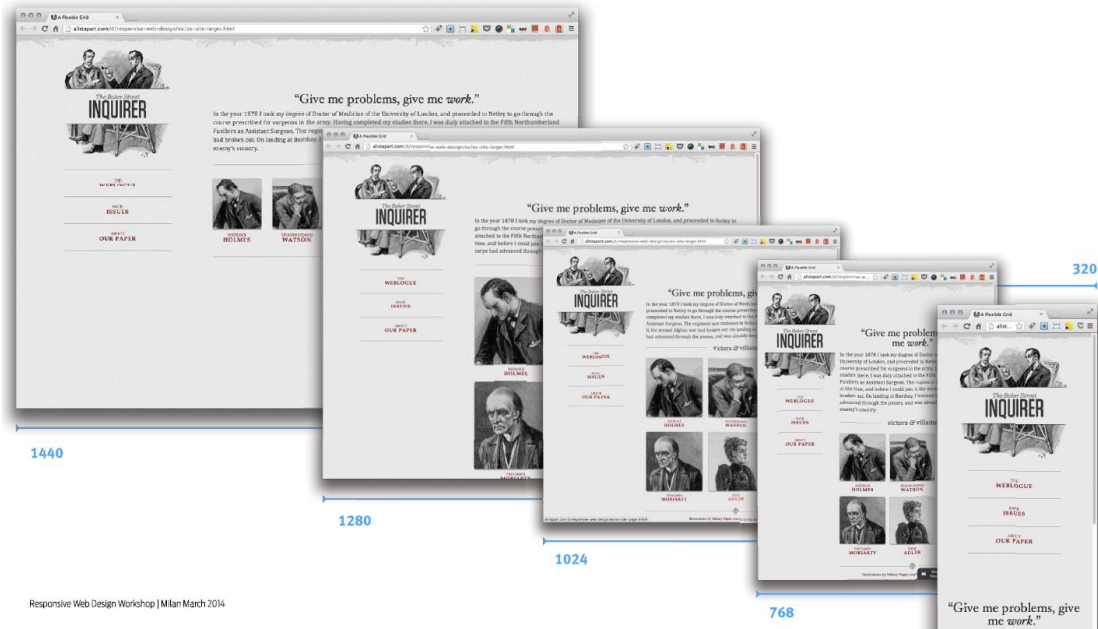
What is Responsive Design?

- Designing website so that it responds to your users environment based on screen-size, platform and orientation.



Media Queries

- Rather than quarantining our content into disparate, device-specific experiences, we can use media queries to progressively enhance our work within different viewing contexts.

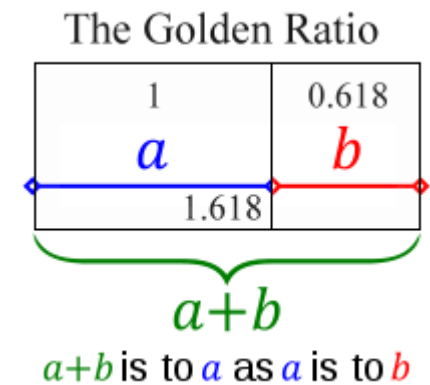
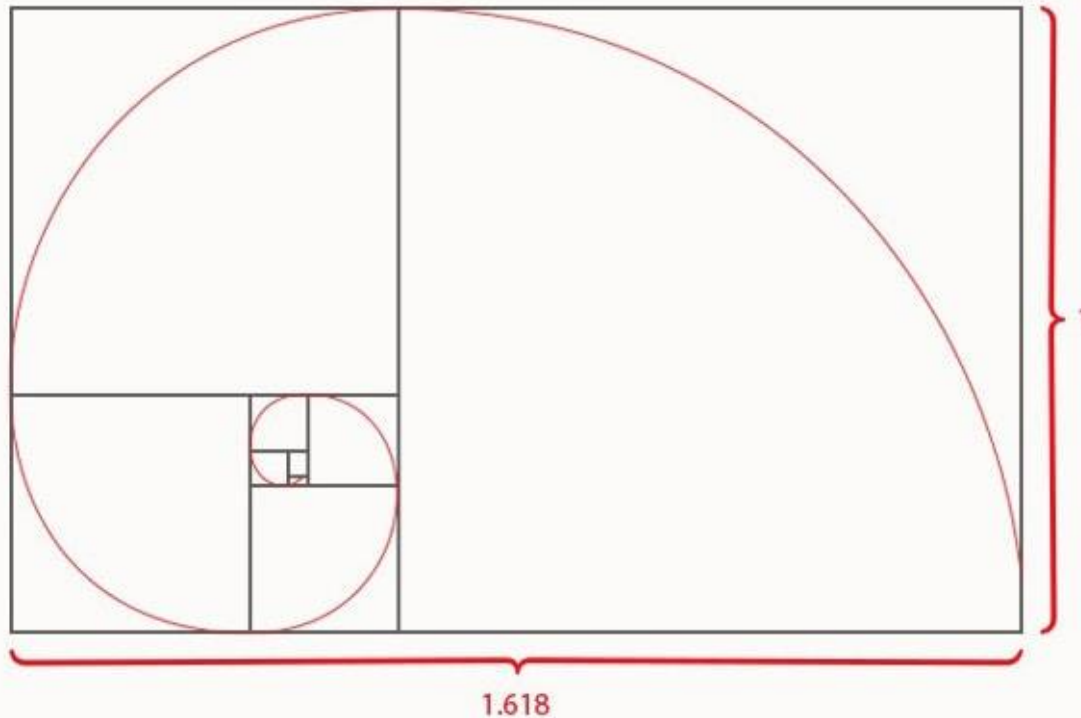


Layout & Alignment

DESIGN PRINCIPLE

The Golden Section

- 1 : 1.618



Credit: <https://www.grappik.com/golden-ratio>

Credit: <https://www.goldennumber.net/google-logo-design-golden-ratio/>

The Golden Section

— PLANTS —

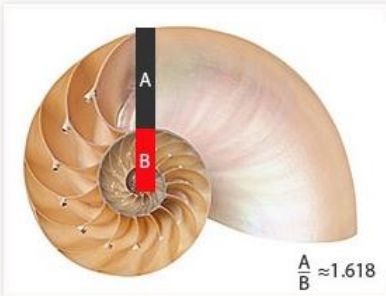


Sunflower

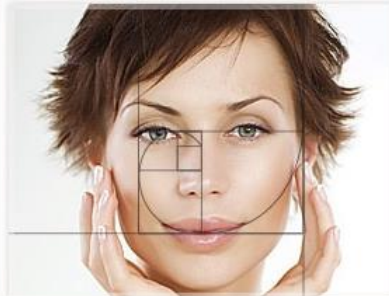


Spiral Aloe

— ANIMALS —

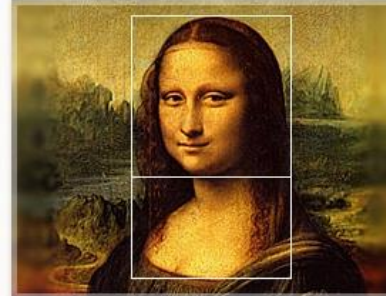


Nautilus Shell



Human Face

— ART —

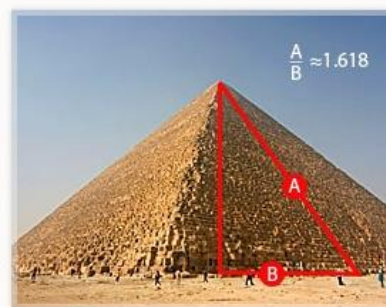


Da Vinci's Mona Lisa



Dali's Sacrament of the Last Supper

— ARCHITECTURE —



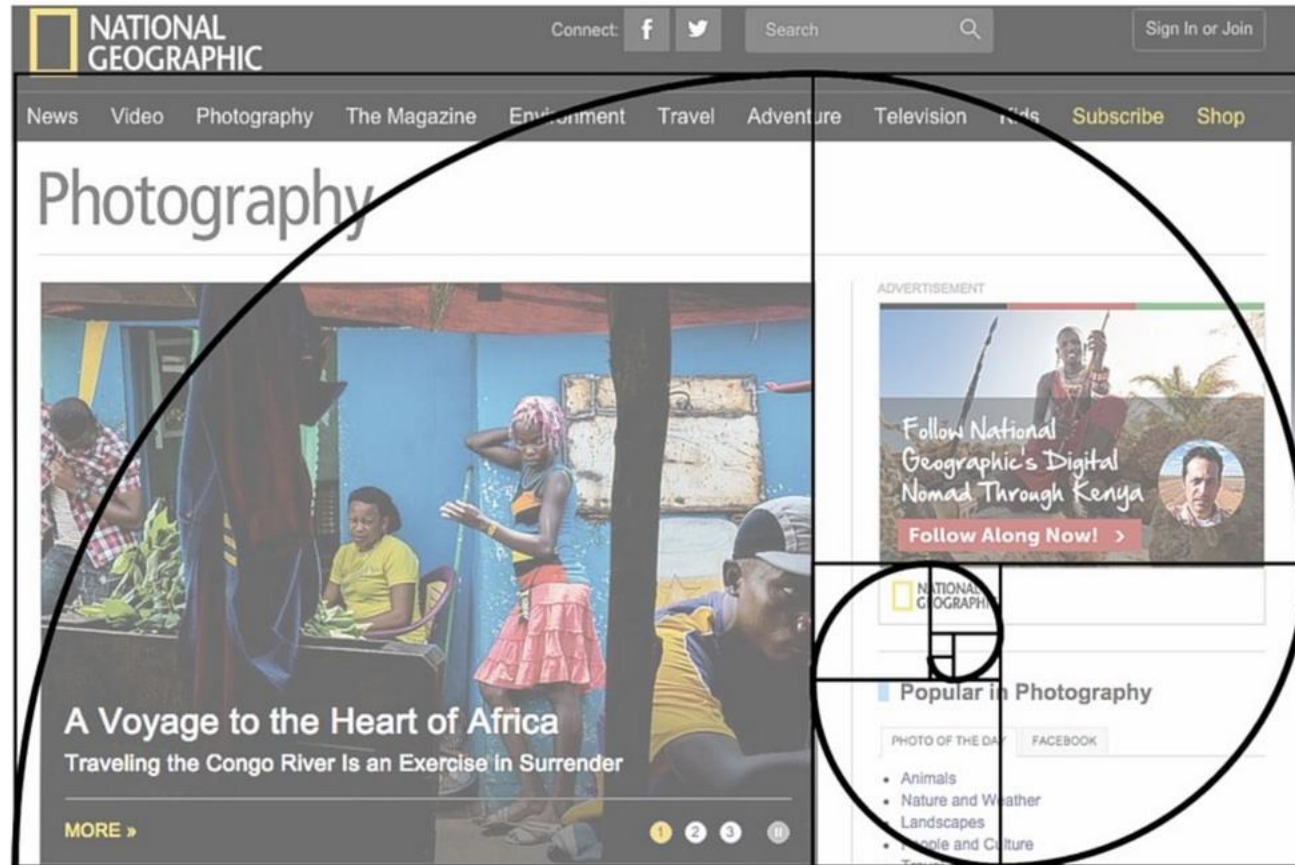
Great Pyramid of Giza



Parthenon

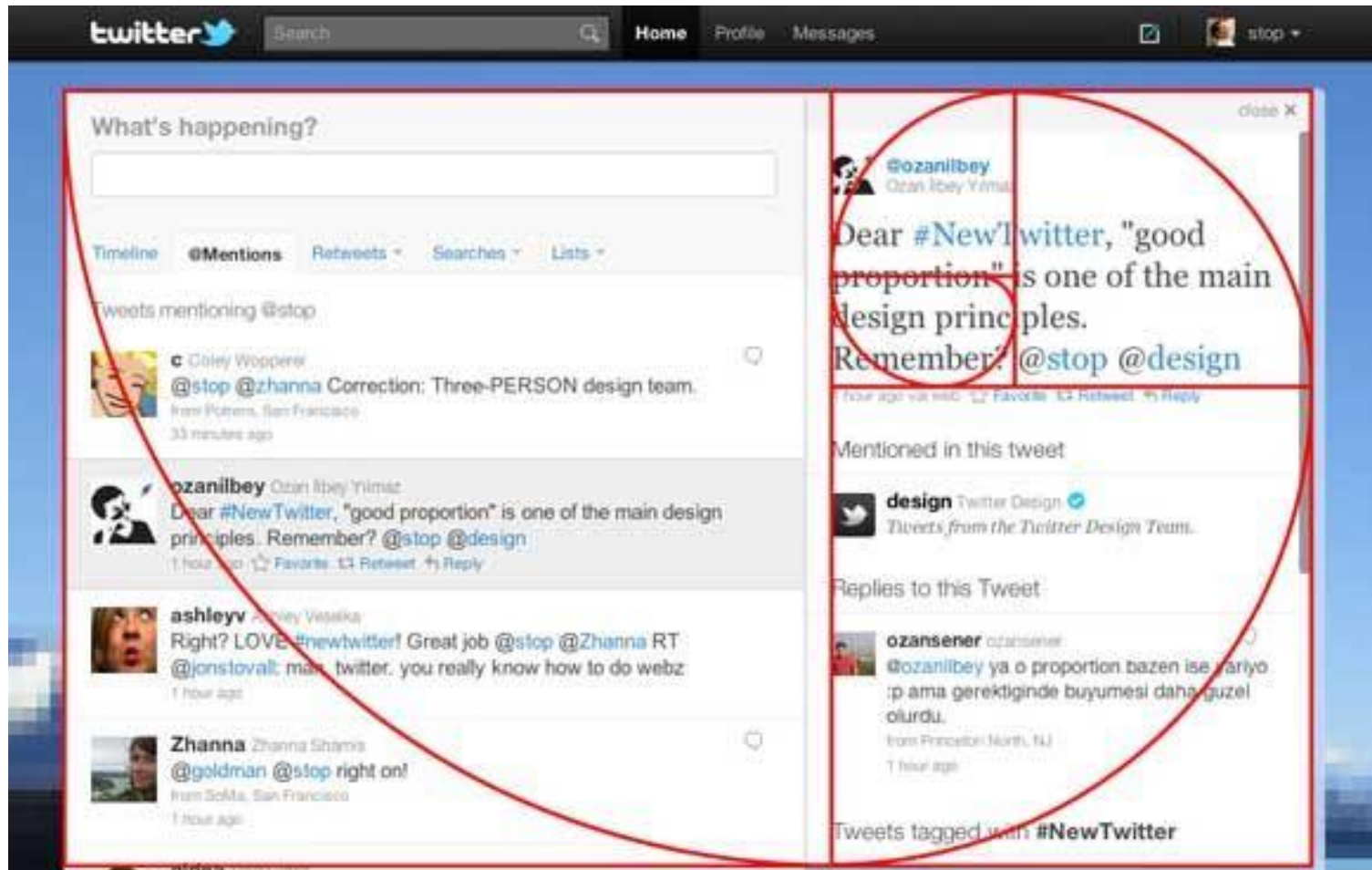
Credit: <https://www.grappik.com/golden-ratio>

The Golden Section



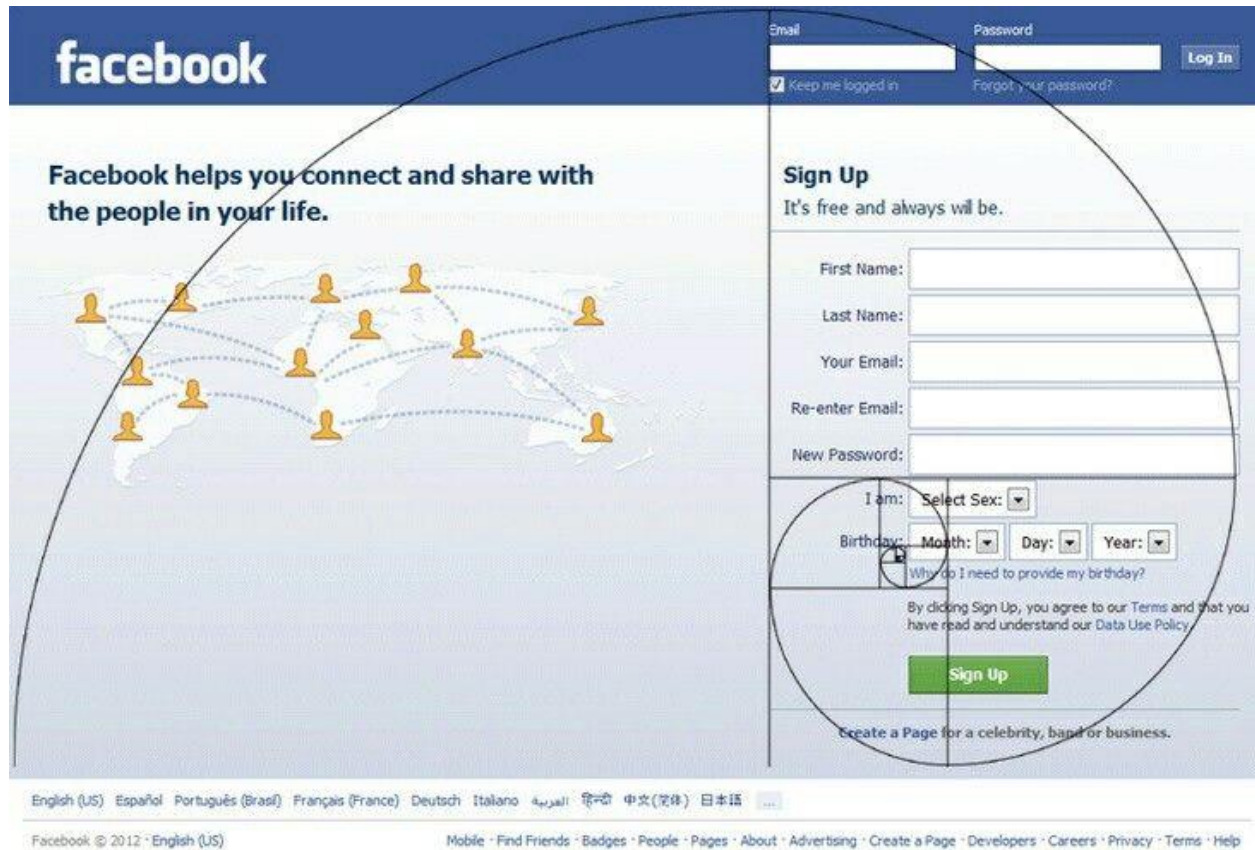
Credit: <https://apiumhub.com/tech-blog-barcelona/golden-ratio-in-web-design/>

The Golden Section



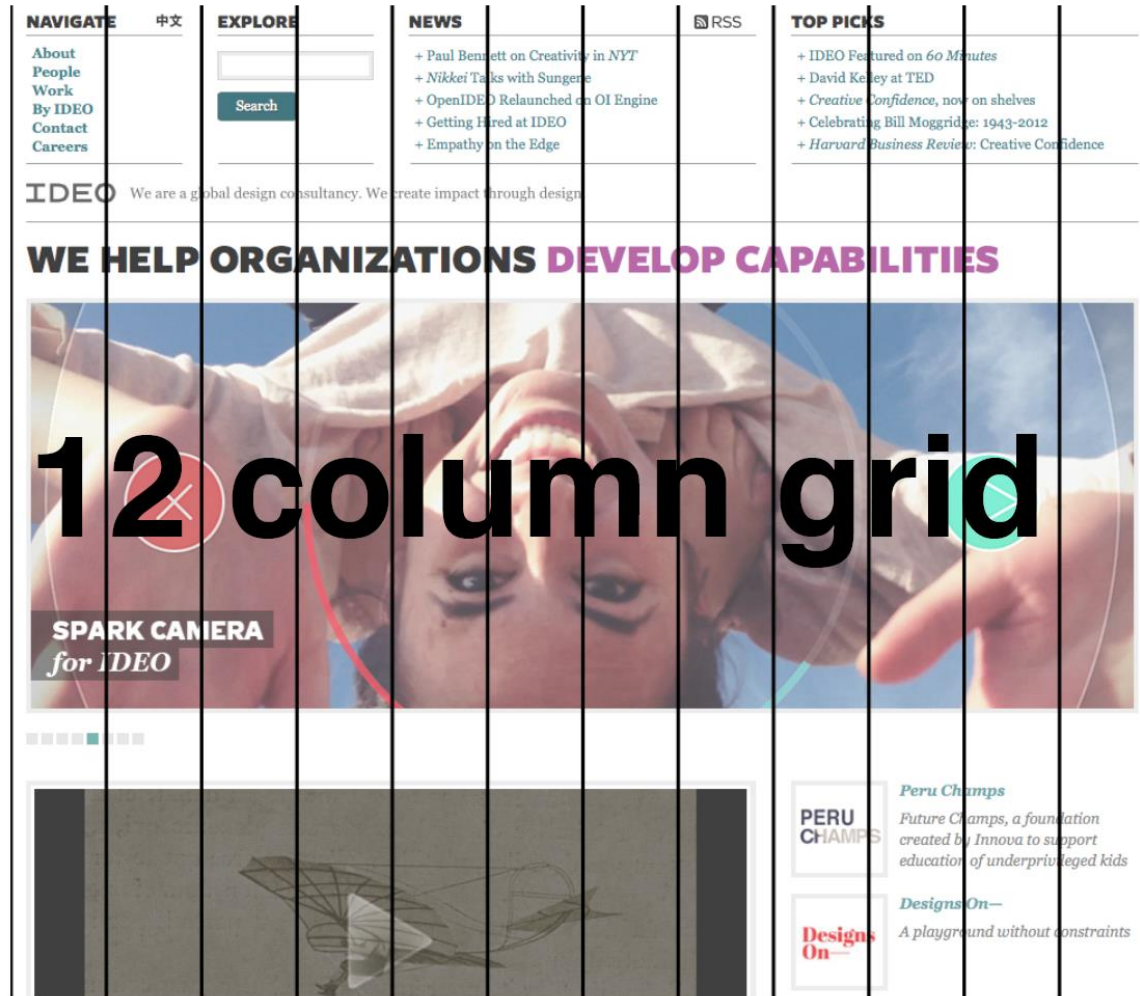
Credit: <https://in.pinterest.com/pin/372250725419571952/>

The Golden Section



Credit: <https://www.facebook.com/>

12 Grid System

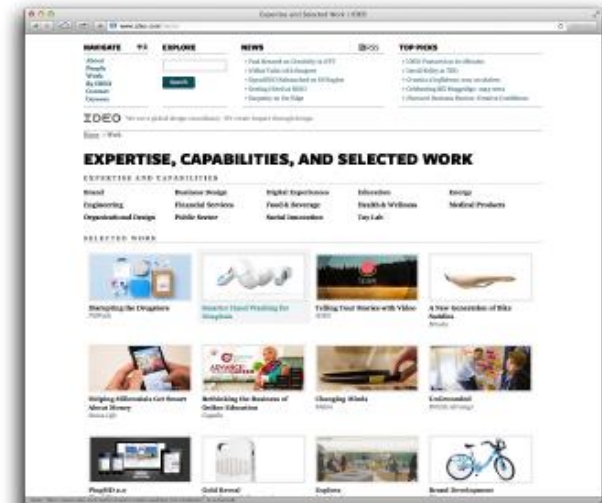
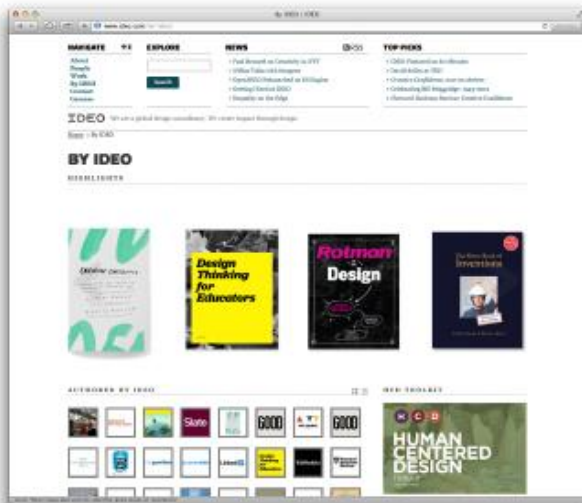


12 Grid System



12 Grid System

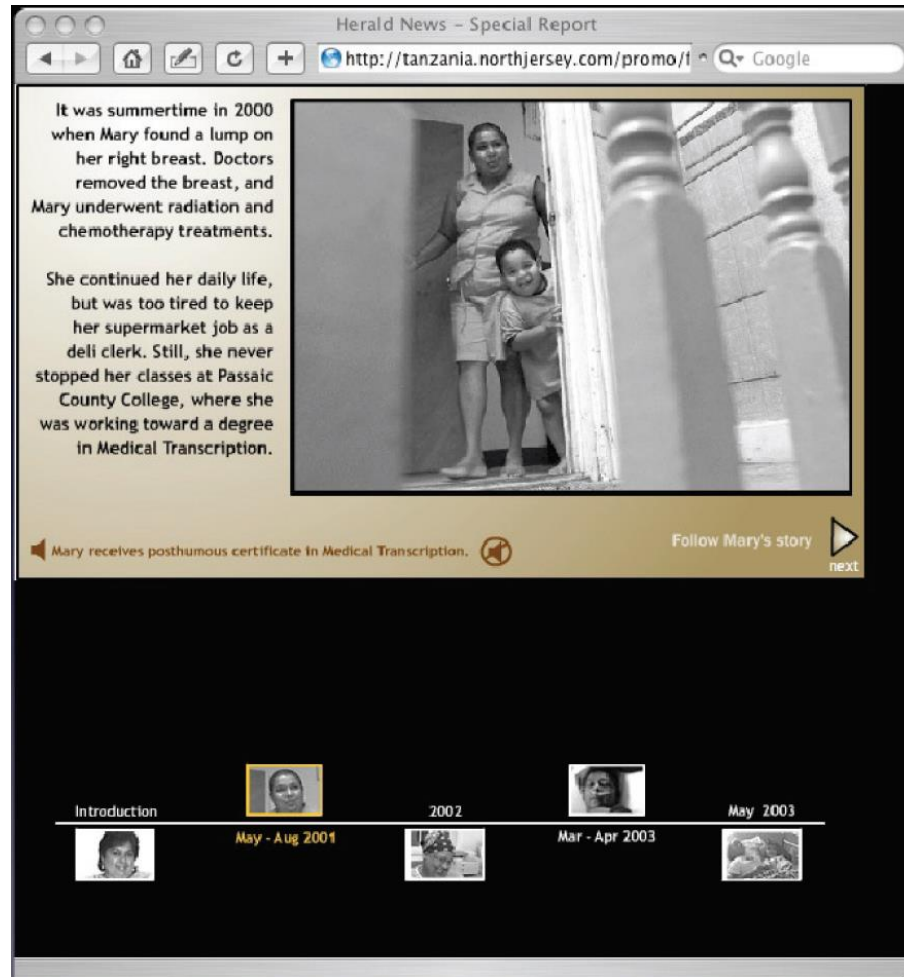
- Why 12?
 - It can achieve many different layouts (1, 2, 3, 4, and 6)



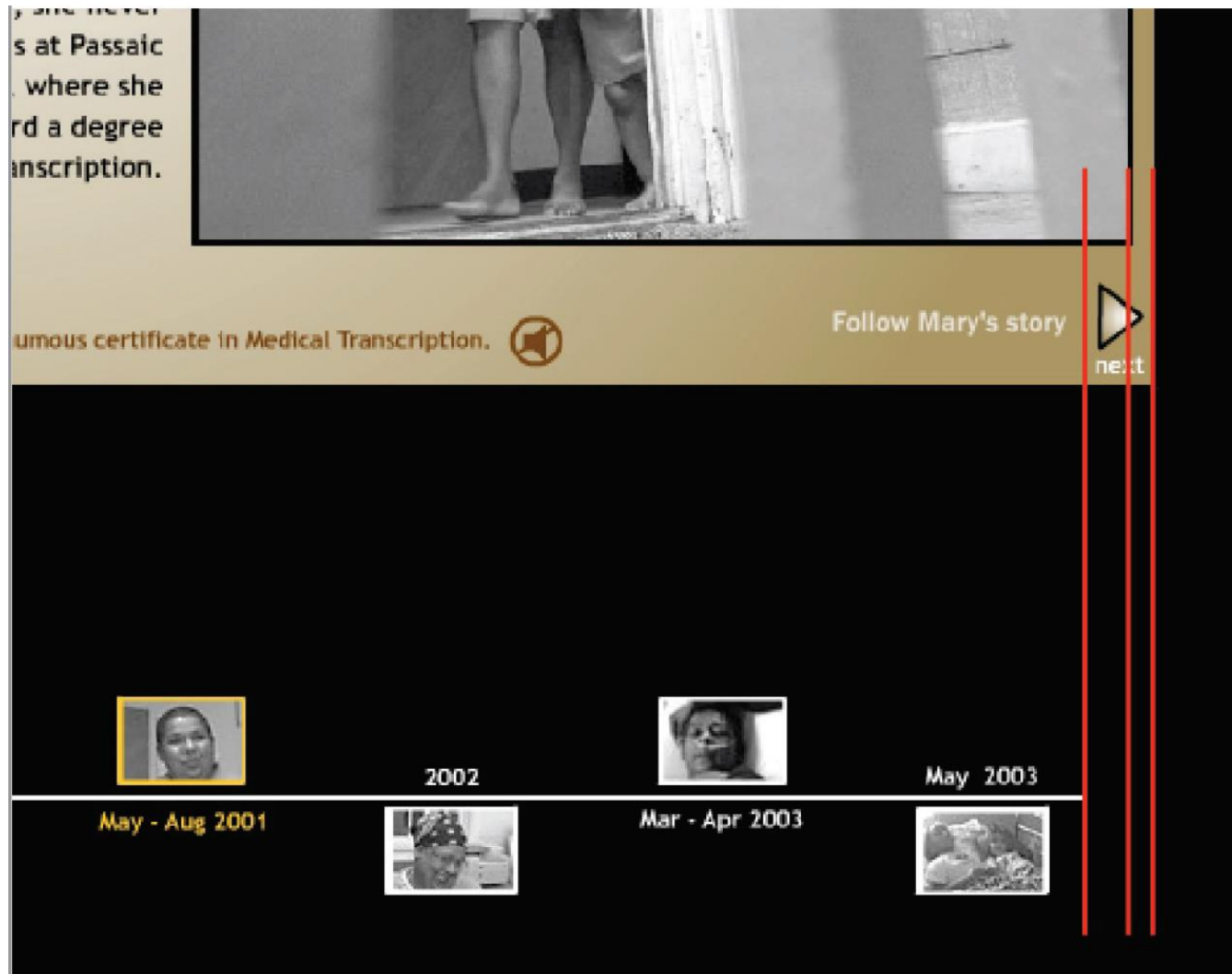
Alignment and Spacing

- Major elements – align edges to one another
- Leave some breathing room, but watch for uncomfortably open spaces
- Elements should have some visual connection to another element on the page/screen
- Don't over-use centering

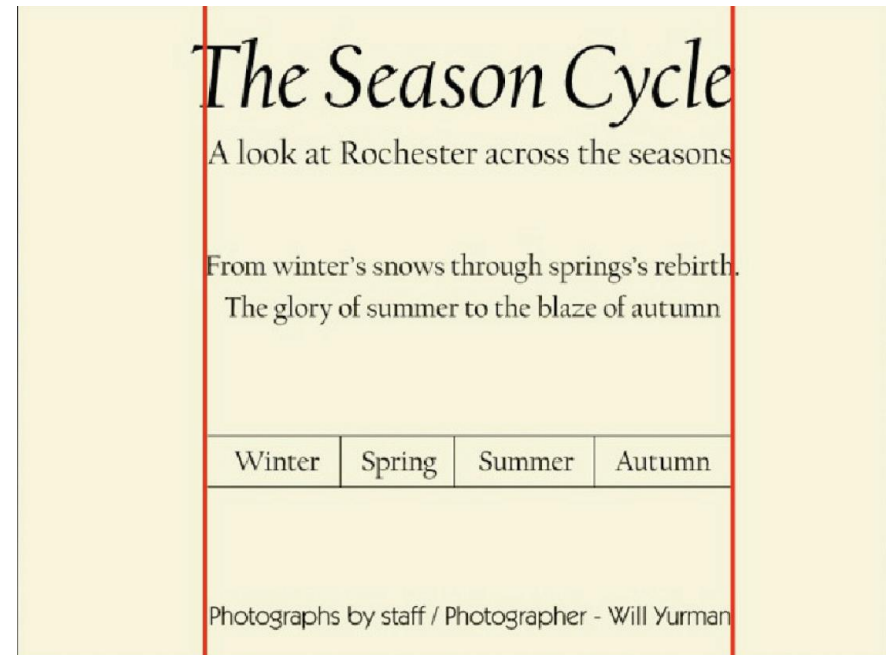
Alignment and Spacing



Alignment and Spacing



Alignment and Spacing

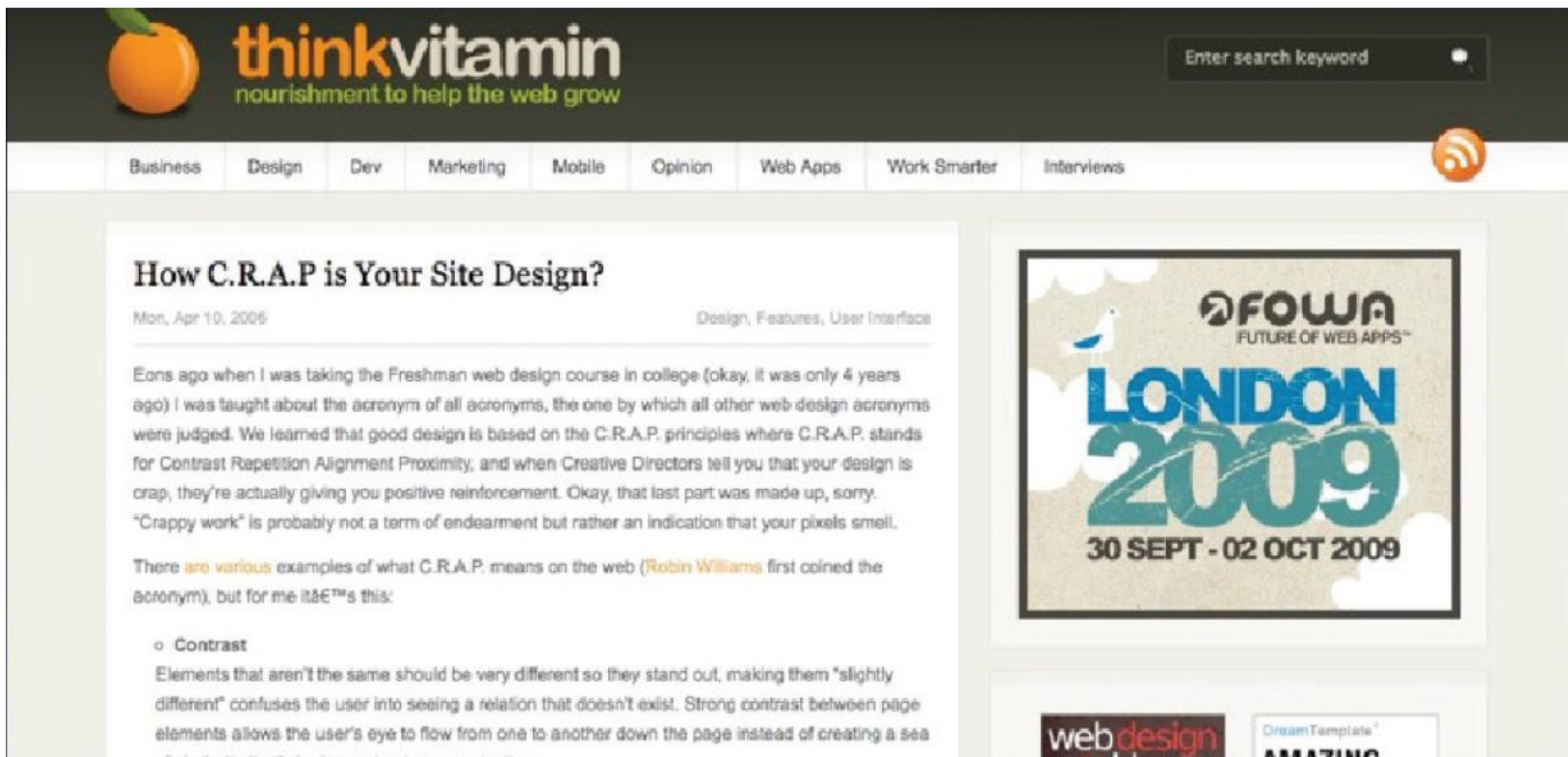


The C.R.A.P. Principle

- Contrast
- Repetition
- Alignment
- Proximity

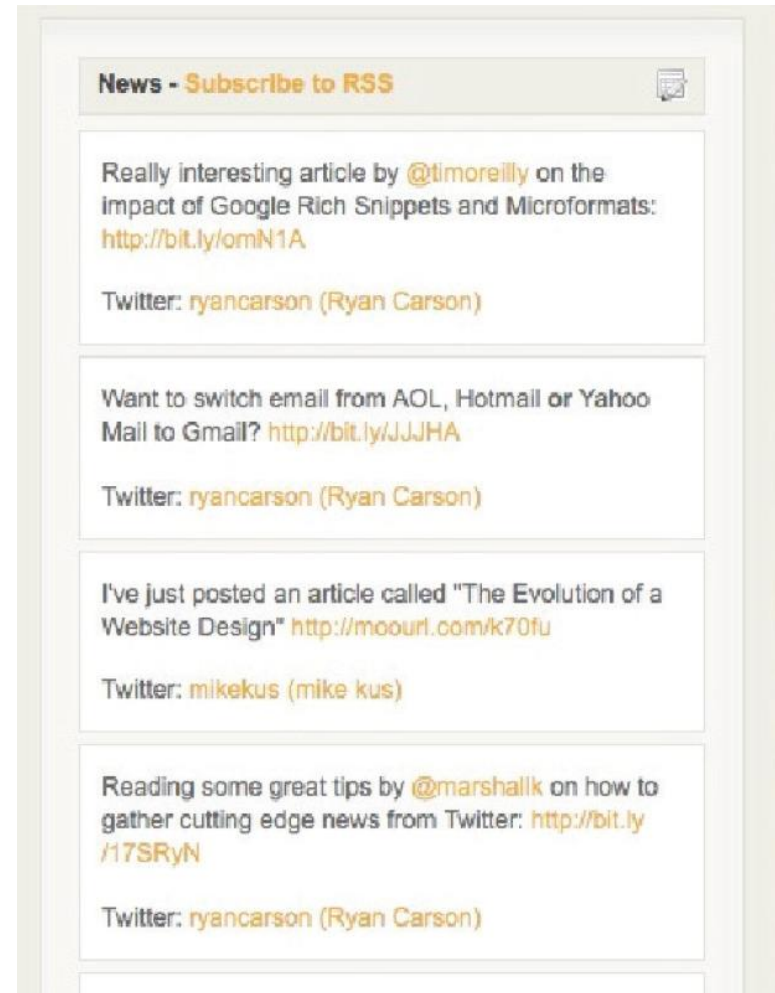
The C.R.A.P. Principle

- Contrast – Major sections differentiated by color



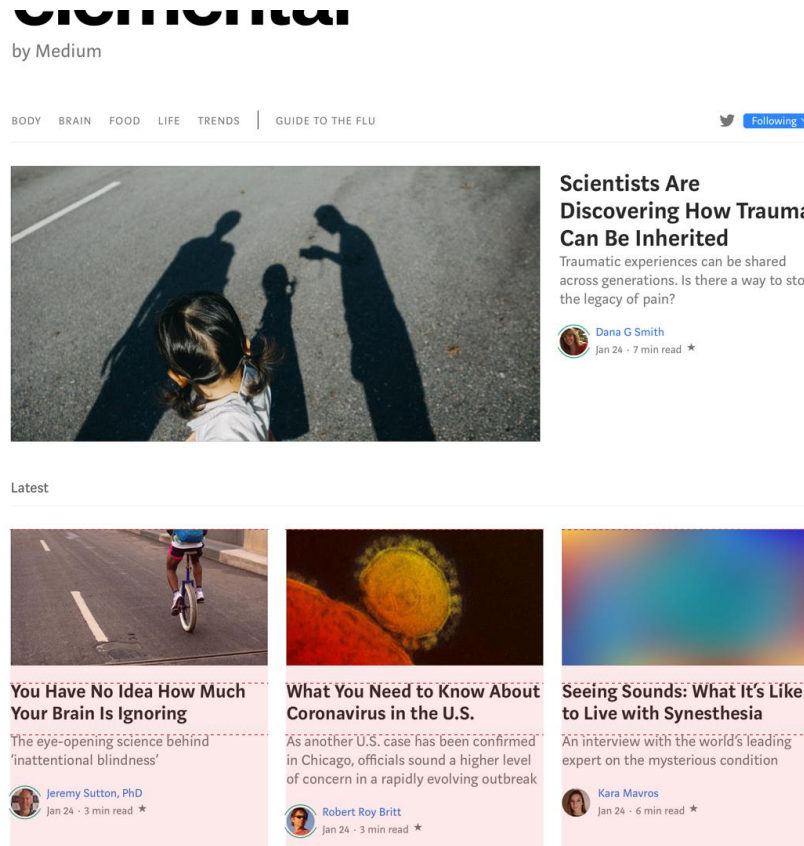
The C.R.A.P. Principle

- Repetition – Repeating design elements down a page reinforces their meaning



The C.R.A.P. Principle

- Alignment – Vertical and Horizontal



Credit: <https://uxdesign.cc/principles-of-ui-design-alignment-dd707e983f29?gi=398d263826c2>

The C.R.A.P. Principle

- Proximity – Related elements closely grouped

ไฮไลต์



‘ตัวอันตรายสายเขียว’รับไม่ไหว เจอใครขับเข้าไล่ยี้ดะ

จับแล้วหมกขังกระบะไล่ยิงรถชาวบ้านบนถนนสายเอเชีย เจ้าตัวอ้างเหตุรถถอยหลัง ปาดหน้า ยี้ดใครไล่ ฆ่าให้เกิดความโศกเศร้าใจอาวรณ์ยิ่งสาหัส



ทูลเกล้าฯขอกรม. 2/2แล้ว ‘สุชาติ-นฤมล-อนุชา’ไม่พลิก

สมชาย แควยวธิน .หนัก80โล ‘บ๊วก’ เหลือลมไฟล้ล้อยเข้า

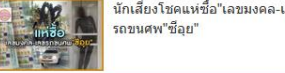


เปิดคลังขั้วพยาน’คดี บอส’ คมขย. ก่อนถูกเฉียวชน



ที่อื่นไม่เชื่อแดนลิซล เชื้อ “ไอ้ไข่” ปรุ่นศรัทธา ครันประทัดคล้าย

บทความไฮไลต์



นักเสี่ยงโชคแห่ซื้อ“เลขมงคล-เลขรทศพ”ซื้อ



"AVA" เอาจ มาแล้ว



"โดยจริง"เกร แก่ไข้อย่างไร?



อาชีพที่รัก"ไอซ์-ปริญญ"เพราะการแสดงไม่มีที่สิ้นสุด

การเมือง บทความการเมือง

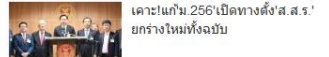


สร.ไม่ฟันธงโควิดระบาดรอบใหม่ จะเกิดจากการชุมนุม

สร.จับตา"เวียดนาม"หลังผู้ป่วยขยายตัวหลายพื้นที่ เผยผู้เชี่ยวชาญ"ต้องคง" เสียงเด็ก ปมการระบาดรอบใหม่เกิดจาก



กมธ.นศ. เล็งเชิญตร-คนเดือนตุลาให้ข้อมูล-ความคิดเห็น



เคาะ'แก๊ง 256'เปิดทางตั้ง'ส.ร.' ยกร่างใหม่ทั้งฉบับ

ข่าวทั่วไทย 9 ศูนย์ข่าว & บทความ



ตั้งเสี่ย

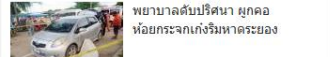
ได้เป็นเจ้าใหม่และครองหัวรั้งแห่งเมือง... (text continues)

ตั้งเสี่ยตามผิดธรรมชาติ อย่าเสียเวลาไปกับกระบวนการยุติธรรม.

เปิดข้อความส่งเสี่ย "หมอมเหยทุทอง" สันหาคลายคดีธรรมชาติอย่าเสียเวลากับกระบวนการยุติธรรม ขอจัดงาน



เปิดใจพจน. ถูกปลดฟ้าผ่า ไร้คำตอบผู้บริหารหัวชนวนเงิน



พบบาลสับปรีสนา ผูกคอ ห้อยกระดกแก๊งมทระดงของ



PANDA motor

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091-709-4777

LINE @pandamotor



Click Here

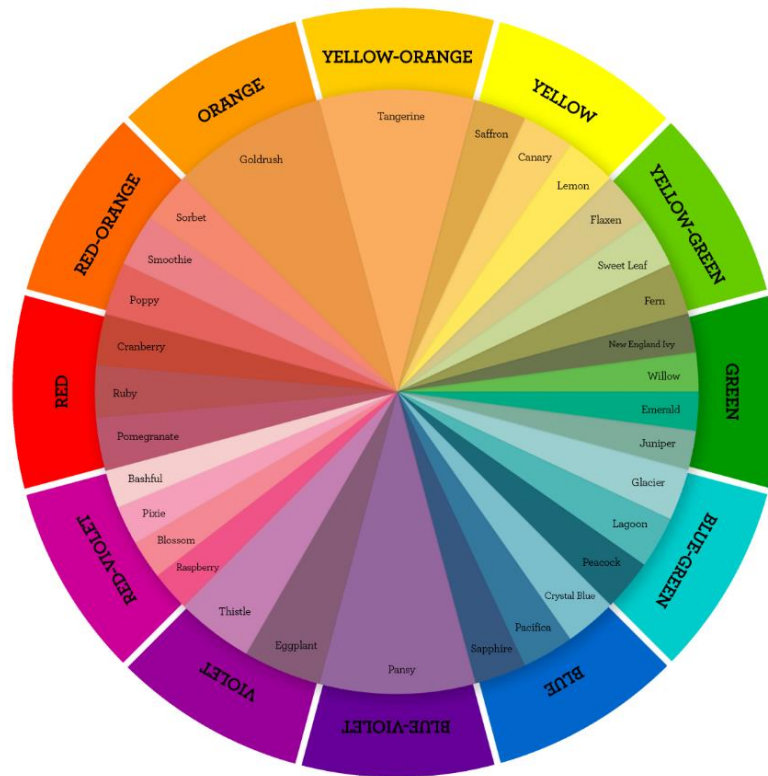


Color

DESIGN PRINCIPLE

Color Theory

- Covers lot of thins but most basic is the interaction of colors in a design through complementation, contrast, and vibrancy.

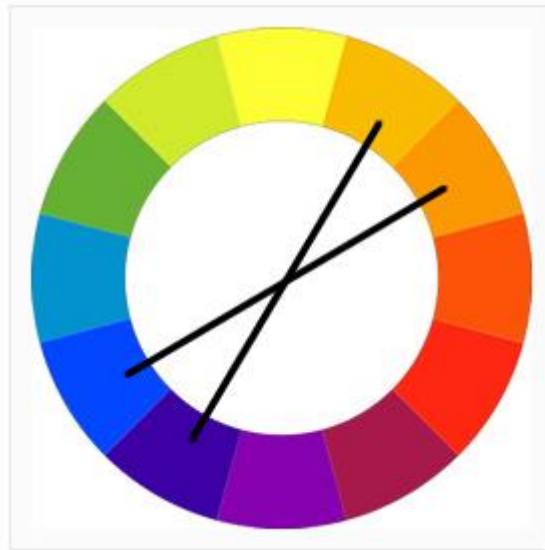


Color Theory

- Complementation – way we see colors in terms of their relationship with other colors. Colors on the opposite end of the color wheel are seen as more visually appealing.



Triadic



Compound



Analogous

Color Theory

- Contrast – reduces eyestrain and focuses user attention by clearly dividing elements on a page.

By choosing stark, complementary colors, text becomes easily readable.

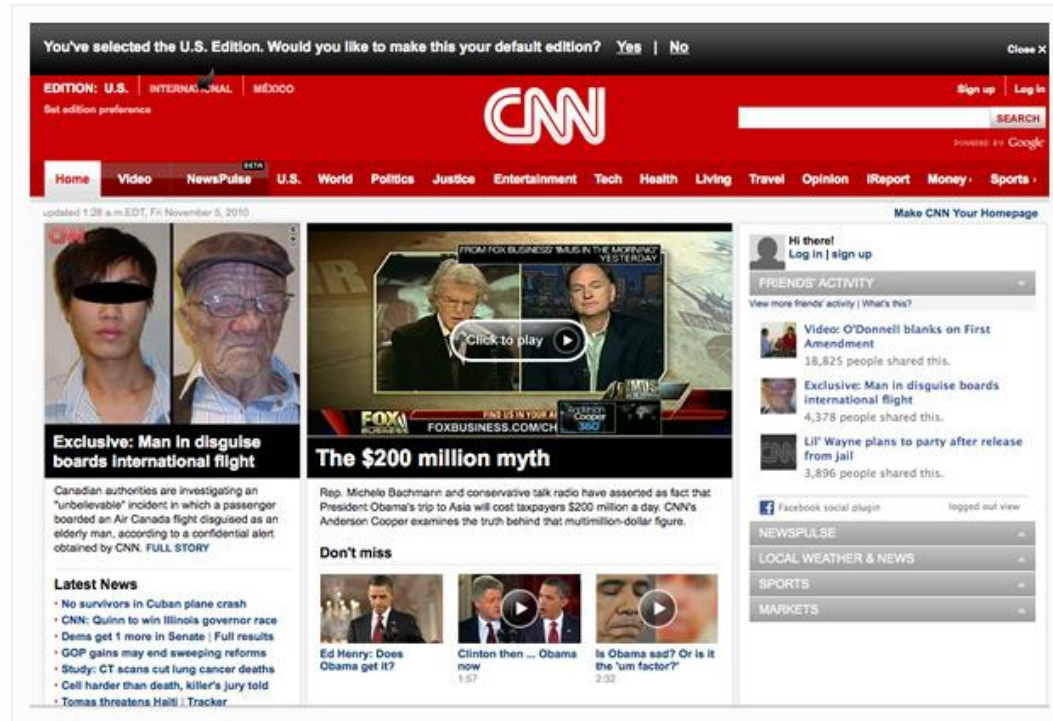
Sometimes, choosing a color scheme in which the text is the brightest element of the design reduces eyestrain by focusing the User's attention.

A lack of contrast between text and its background will drive your users insane. Their eyes don't know which color to focus on, which will almost instantly strain their eyes.

A complementary color scheme does not necessarily mean the contrast between text and background is strong enough. Sometimes, one of your colors will be too bright in comparison to the other, which will lead to eventual eyestrain.

Color Theory

- Vibrancy – dictates the emotion of your design. Brighter colors lead the user to feel more energetic while darker colors relax users, allowing their mind to focus on other things.



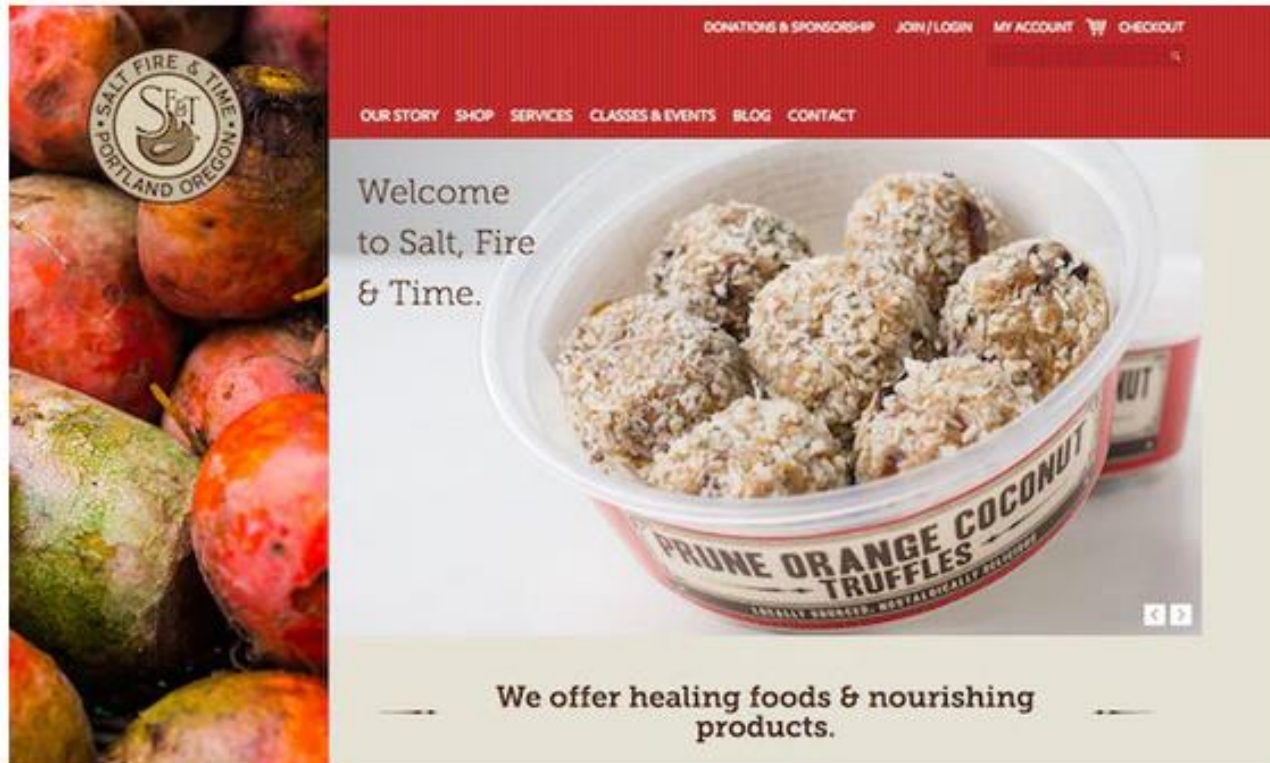
Color Theory

- Three types of colors:
 - Primary colors - cannot be created by mixing other colors together.
 - Secondary colors - are formed by mixing two primary colors: blue and yellow make green, red and yellow make orange.
 - Tertiary colors - are derived by combining primary and secondary colors: blue-green or red-orange.
- Web designers use these types of colors to evoke feelings when designing a blog.

The Psychology of Color

- **Red** can actually increase a person's heart rate and cause them to breath faster.
- **When to use red** - to draw attention to something, or to create excitement (e.g., food, fashion, entertainment, sports, marketing, advertising, emergency services, and health care).
- **When to avoid red** - Don't overuse it! Too much excitement can be a bad thing. Red is generally not suitable for luxury goods, nature-related content, or professional websites/services.

The Psychology of Color

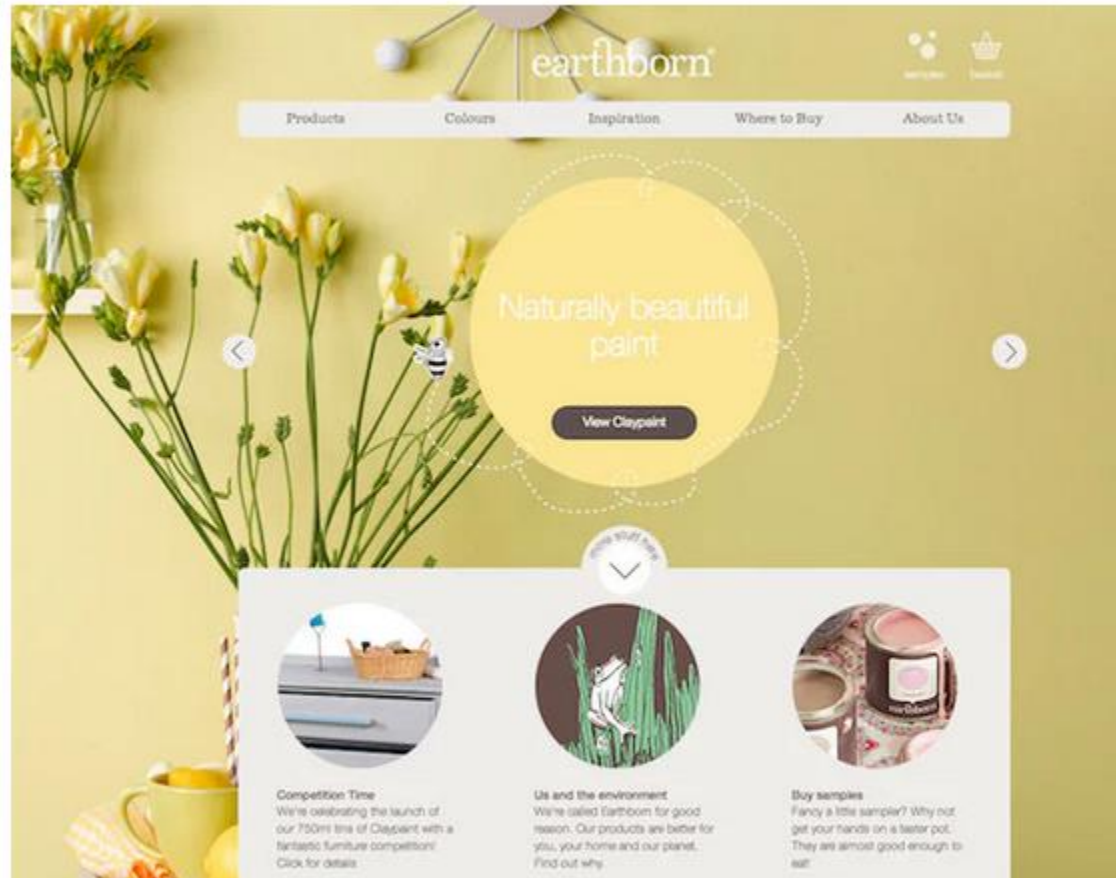


Salt, Fire & Time

The Psychology of Color

- **Yellow** is associated with competence, happiness, cheer, optimism, and youth.
- **When to use yellow** - to energize people or to create a sense of happiness. Use soft, light yellows for a calmer happy feeling.
- **When to avoid yellow** - Yellow can quickly become overpowering. It can strain the eyes. Again, use it sparingly. Too much yellow or the wrong hue can feel cheap or spammy.

The Psychology of Color

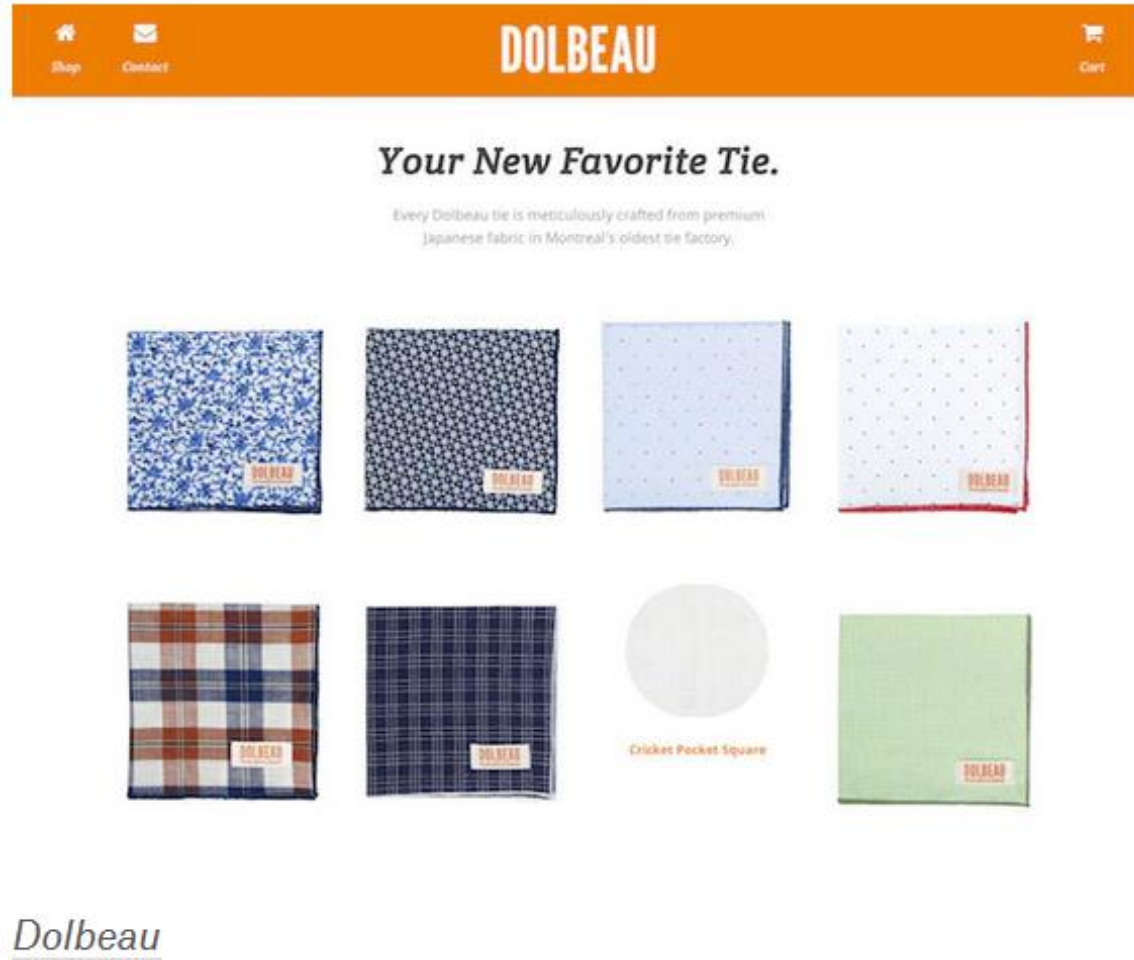


Earthborn Paints

The Psychology of Color

- **Orange** is an energetic and vibrant color often associated with fun, happiness, energy, warmth, ambition, excitement, and enthusiasm. It can also be used to communicate caution.
- **When to use orange** - Draw attention to your call to action (subscribe, buy, sign up), clearance, sales, or other content you want to make sure people notice.
- **When to avoid orange** - Although orange is a bit less intense than red, it can still quickly become overpowering. Don't overdo it!

The Psychology of Color



The Psychology of Color

- **Green** has a harmonizing, balancing effect. It is associated with growth, health, nature, wealth, money, calmness, masculinity, generosity, fertility, envy, good luck, peace, harmony, support, and energy.
- **When to use green** - Green is the easiest color for the eye to process. Use it to create a relaxing, calming effect or to represent new beginnings, nature, or wealth.
- **When to avoid green** - It's less appropriate for luxury goods, tech, or content geared toward adolescent girls.

The Psychology of Color

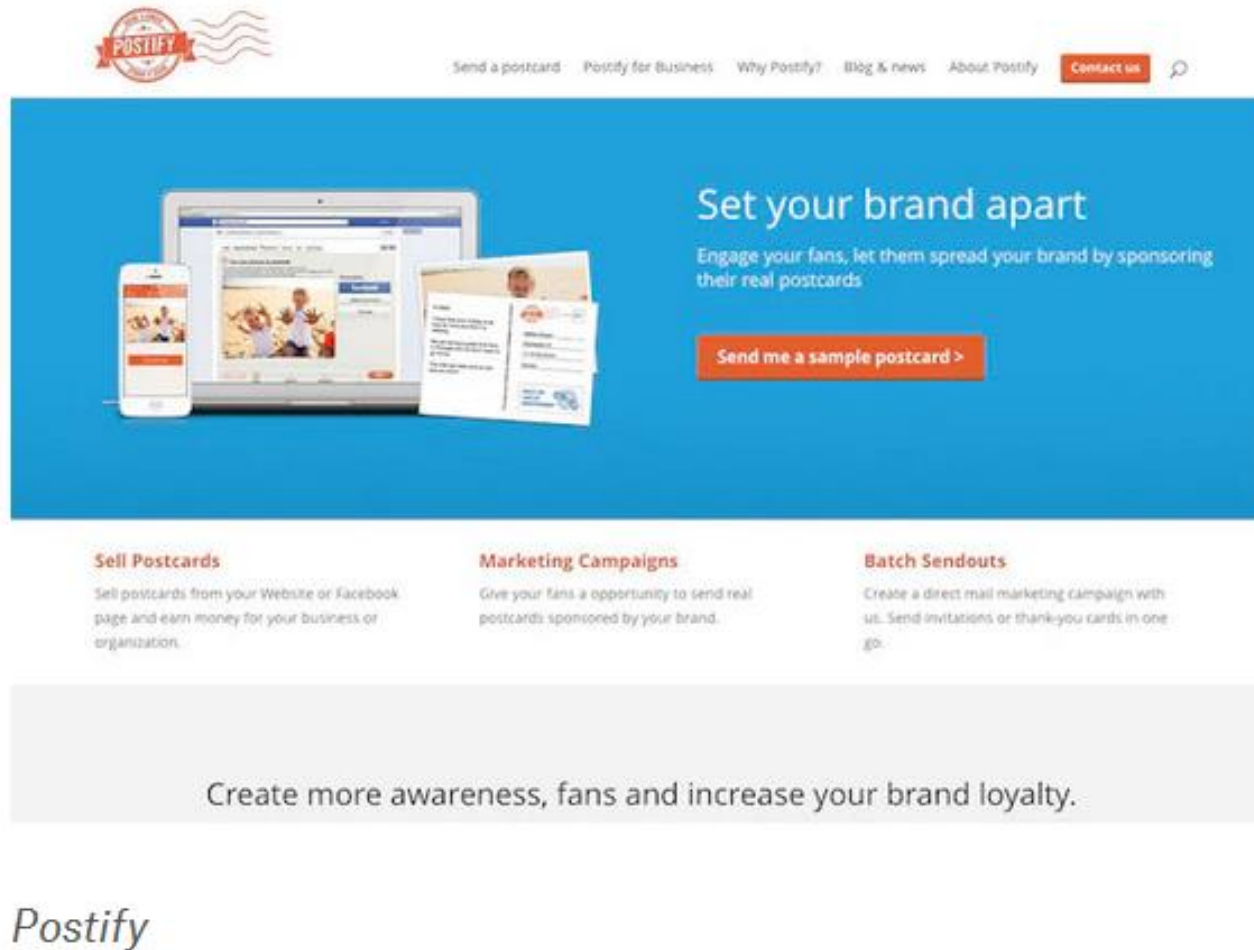


Geo Fun

The Psychology of Color

- **Blue** is associated with masculinity, competence, quality, calmness, dependability, steadfastness, wisdom, loyalty, strength, productivity, trust, and security. Bright blues can also be refreshing and energizing.
- **When to use blue** - is often used by large corporations and banks because it's non-invasive and associated with dependability. It's good for health care, dental, high-tech, medical, science, government, legal, and utilities.
- **When to avoid blue** - Using certain shades of blues (on the darker end of the spectrum), or using too much blue can make your website feel uncaring and cold. Blue may also curb appetite, so be careful using it with food-related content.

The Psychology of Color



The screenshot shows the Postify website homepage. At the top left is the Postify logo, which includes a red circular seal with the word 'POSTIFY' and a wavy line. To the right of the logo is a navigation menu with links: 'Send a postcard', 'Postify for Business', 'Why Postify?', 'Blog & news', 'About Postify', and a red 'Contact us' button. Below the navigation menu is a large blue banner. On the left side of the banner is an image of a smartphone, a laptop, and a stack of postcards. On the right side of the banner, the text reads 'Set your brand apart' in white, followed by 'Engage your fans, let them spread your brand by sponsoring their real postcards'. Below this text is a red button that says 'Send me a sample postcard >'. Below the blue banner are three columns of text. The first column is titled 'Sell Postcards' and describes selling postcards from a website or Facebook page. The second column is titled 'Marketing Campaigns' and describes giving fans a chance to send real postcards sponsored by the brand. The third column is titled 'Batch Sendouts' and describes creating a direct mail marketing campaign. At the bottom of the page is a light gray box with the text 'Create more awareness, fans and increase your brand loyalty.' and the Postify logo.

Postify

The Psychology of Color

- **Purple** is associated with royalty. It can be used to communicate creativity, imagination, authority, sophistication, power, wealth, prosperity, mystery, wisdom, and respect.
- **When to use blue** - use dark purples to create a sense of luxury and wealth, and use light purples for spring and romance. Purple can also be great for beauty products (specifically anti-aging), astrology, massage, yoga, healing, spirituality, and content related to adolescent girls and feminine brands.
- **When to avoid blue** - Purple can be soothing and calming which generally makes it a poor choice for grabbing people's attention. The use of darker, deeper purples can make your site feel aloof or distant.

The Psychology of Color



Awesomewall

The Psychology of Color

- **Brown** is a warm, natural color associated with earth, ruggedness, reliability, stability, friendship, and nature.
- **When to use brown** - can be used to stimulate appetite (think of commercials for coffee and chocolate), making it suitable for food-related content. It can also be a good fit for real estate, animals, veterinary, and finance. Brown is also typically better for backgrounds.
- **When to avoid brown** - can be a bit boring or overly conservative. It's generally not suitable for grabbing people's attention and should not be used for call to action items.

The Psychology of Color

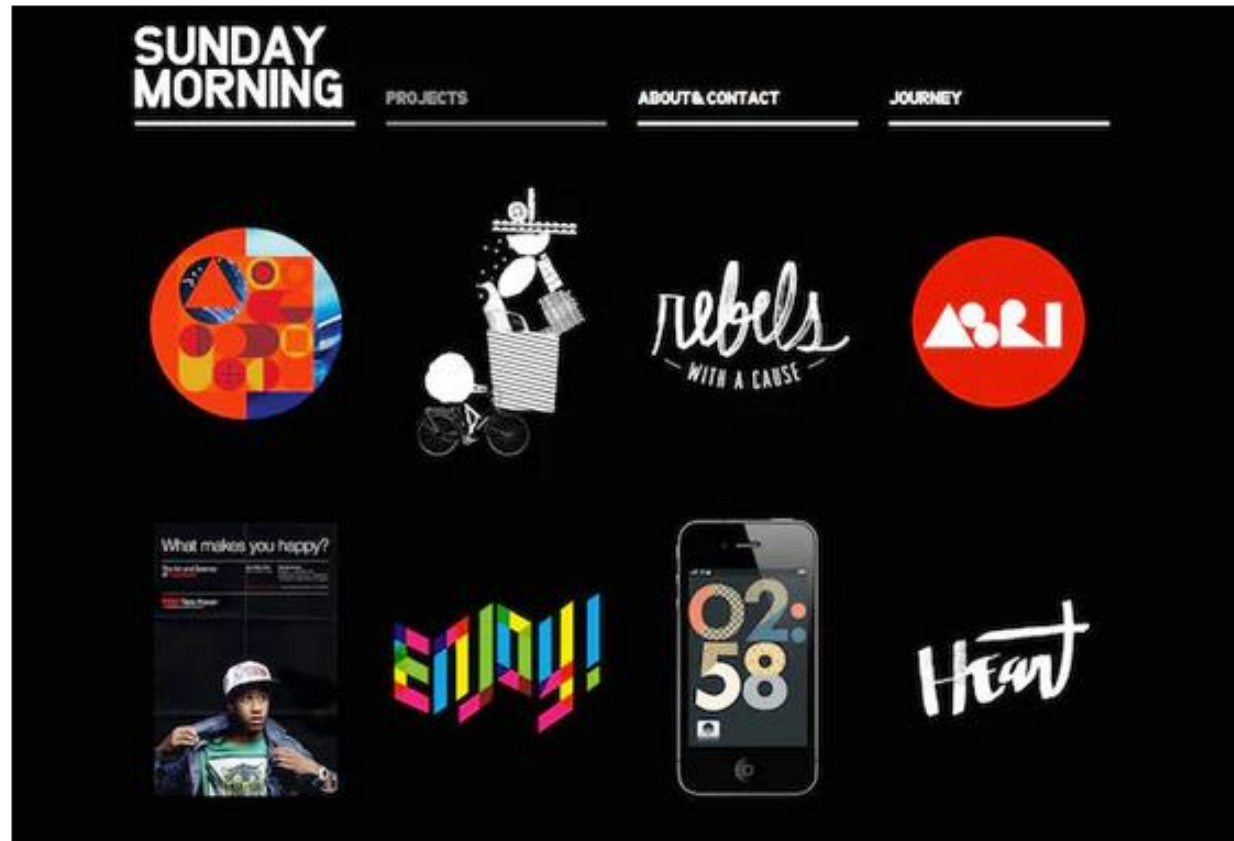


Concrete Matter

The Psychology of Color

- Black is a strong color often associated with sophistication, elegance, authority, power, sleekness, stability, strength, formality, and intelligence. It can also symbolize death, mystery, evil, and rebellion.
- **When to use black** - Depending on the colors used with it, black can be elegant and traditional, or modern and edgy. Black can be great for luxury goods, fashion, marketing, and cosmetics.
- **When to avoid black** - Too much black can quickly become overwhelming. Black can also feel menacing or evil, making people feel uncomfortable or even afraid.

The Psychology of Color



Sunday Morning

The Psychology of Color

- **White** is associated with purity, cleanliness, virtue, happiness, sincerity, and safety.
- **When to use black** - is associated with doctors, nurses, and dentists which makes it great for websites related to the health care industry. It can also work for high-tech and science sites. When paired with black, gold, silver, or grey, white can also be great for luxury goods.
- **When to avoid black** - Since the effects of white depend almost entirely upon the other colors in the design, it can theoretically be used for any type of website.

The Psychology of Color



Satchmi

The Psychology of Color

- Gray is associated with formality, professionalism, sophistication, practicality, timelessness, and strong character.
- **When to use gray** - it's great for professional websites, luxury goods, or to create a balancing, calming effect.
- **When to avoid gray** - Certain shades of grey may feel dull and detached, or even cold. Grey is not ideal for grabbing people's attention.

The Psychology of Color

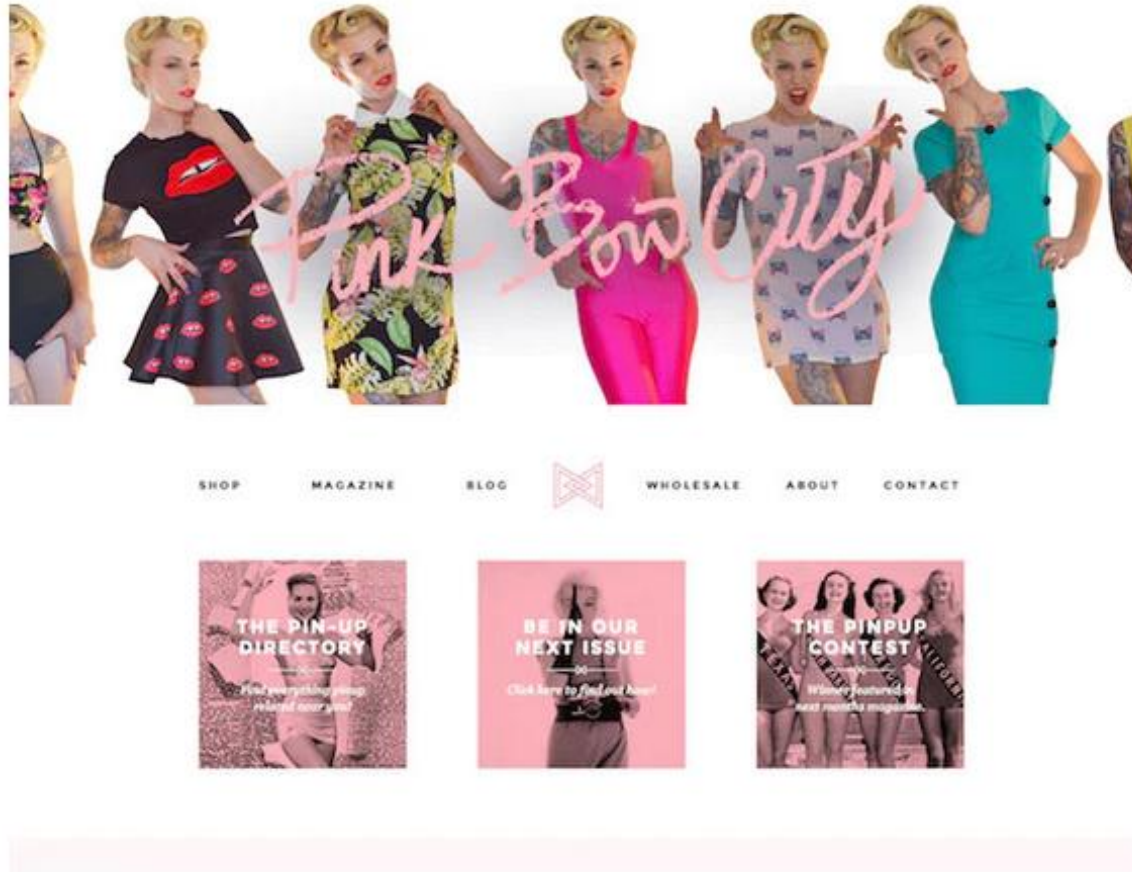


Bird Boutique

The Psychology of Color

- **Pink** represents sophistication, sincerity, romance, and love. It does not have the violent, angry connotations of red, and it can be quite soothing and gentle.
- **When to use pink** - is great for feminine products or sites with content specifically geared towards women and young girls.
- **When to avoid pink** - Bright pinks can be gaudy and light pinks can feel too sentimental or sweet for some sites.

The Psychology of Color



Pink Bow City

7 Rules for Website Color Schemes

- Research shows that 62-90% of snap purchase decisions are driven purely by the perception of colors.
- An average user has an attention span of 8 seconds. That's how long you have to convince them to continue browsing your website.
- Choosing the right color scheme for your web design will help you engage the users with color accents, sharp navigational elements, and memorable branding.

7 Rules for Website Color Schemes

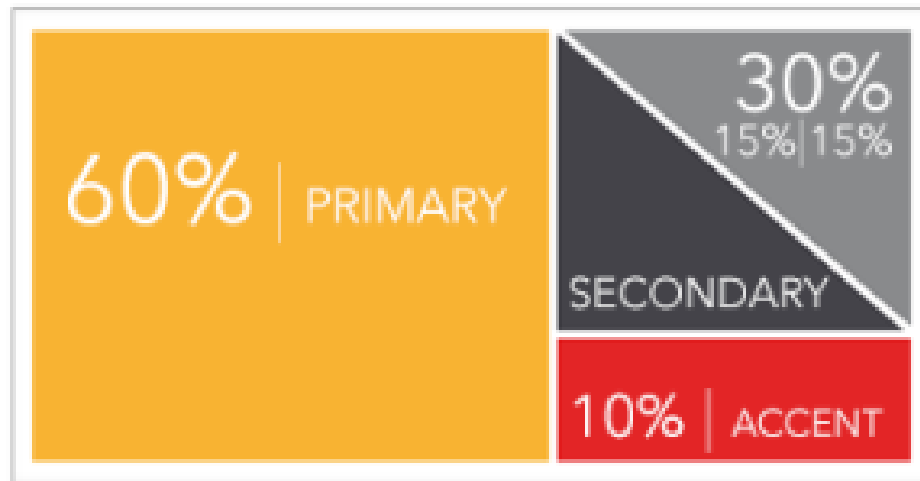
1. Find Your Primary Color

- If you have a logo or any existing branding in place, there's probably one main color that's dominating the designs. This is the color that your brand will be associated with – your primary brand color.

7 Rules for Website Color Schemes

2. Decide on The Number of Colors

- It's typically recommended to use a three-color combination.
- There's also an old designer's rule that can be helpful when applying your color scheme to the website design. It's known as the **60-30-10 rule**.



7 Rules for Website Color Schemes

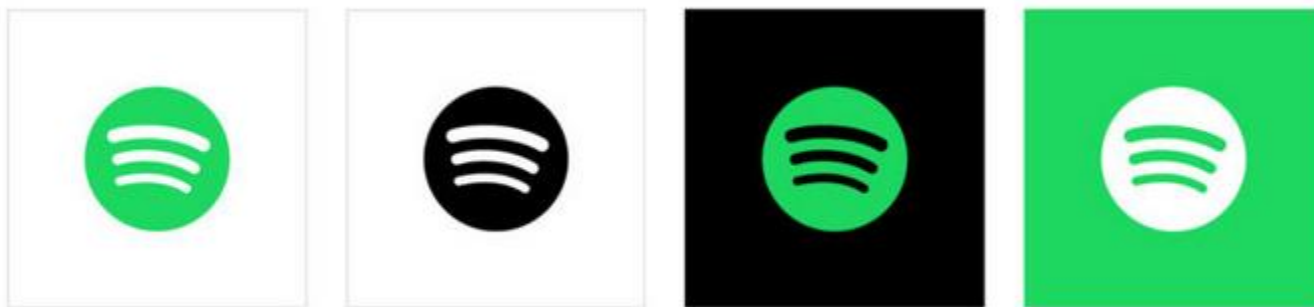
3. Use Secondary Colors When Needed

- Sometimes a design will require more than just a few primary colors. This is especially true for websites that are packed with content and landing pages – products, enablement materials, downloadable assets, etc.
- The ability to visually separate headings, sidebars, captions, buttons, and other elements increase the conversion rate and improve the user experience.
- Sometimes, it's worth considering adding different shades and tints of your main colors instead of looking for additional ones.

7 Rules for Website Color Schemes

4. Don't Forget Your Neutral Colors (White, Black Gray)

- You will need neutral colors for text, to create contrast for important elements, or as background colors.
- The function of neutral color in modern web design is much like that of white space. It's a good idea to have one dark and one light neutral color for different use cases.



7 Rules for Website Color Schemes

5. Add Primary, Secondary and Neutral Colors In The Following Places:

- **Primary colors** go to the “hot spots” on your web page. You should use these bold, vibrant colors to attract users’ attention and prompt them to take action. CTA buttons, headlines, benefits icons, download forms, and other important information should be highlighted using primary colors.
- **Secondary colors** are used to highlight the less important information on the website, such as secondary buttons, subheadings, active menu items, backgrounds, or supporting content like FAQs and testimonials.
- **Neutral colors** will most likely be used for text and background but could come in handy in particularly colorful sections of the site, just to help tone it down and refocus the eye.

7 Rules for Website Color Schemes

6. Go Through The Process Several Times

- When you finally devise your color palette, put it to the test in various scenarios.
 - Will it work in print?
 - What if you decide to start a merchandise line?
 - Does it have an effect on visitors you expected it to have?

7 Rules for Website Color Schemes

7. Brainstorm When Needed

- Beautiful color schemes are not that hard to come by when you know where to look. Here's a list of helpful tools that will guide you in your search for the best website color combinations.
 - <https://mycolor.space/>
 - <https://color.adobe.com/create>
 - <https://paletton.com/#uid=1000u0klIIlaFw0g0qFqFg0w0aF>
 - <https://coolers.co/>
 - <https://www.colorsfire.com/>
 - <https://www.canva.com/colors/color-wheel/>

Exercise 4.1

- Visit your most favorite website regarding the design and discuss following issues:
 - Why do you like it?
 - What design principles does the website follow?
- Find bad example of website regarding the design and discuss why the design is inappropriate.